Fisher's Sharp Null: A Four-Student Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 1 Hour 4 and Day 2 Hour 2 (randomization inference) and Lab 1.

Step 1. The question

Four students, Ahmed, Bader, Dana and Reem, take the same short math test. Claim: an energy drink before the test makes a student finish faster.

What we want, defined before any data **estimand: the ATE**: the average, over the four students, of each one's time with the drink minus the same student's time with water.

$$\text{ATE} = \text{mean over students of } [\ Y_i(\text{energy drink}) - Y_i(\text{water})\ ]$$

The same average over only the two students who got the drink is the ATT **estimand (ATT)**. Under randomization ATE and ATT agree on average.

Step 2. The sharp null hypothesis

The sharp null: **the drink does nothing for anyone**. Each student's time is the same with the drink or with water. For every student i:

$$Y_i(\text{energy drink}) = Y_i(\text{water}), \text{ so } \tau_i = Y_i(\text{energy drink}) - Y_i(\text{water}) = 0 \text{ for every } i$$

"Sharp" means the claim is about every student, not only the average. If it is true, both potential outcomes are known for everyone, because they are equal. That is what makes it testable with four people.

Step 3. The design

Coins pick 2 of the 4 at random for the drink (treatment); the other 2 get water (control). Record minutes to finish.

Step 4. The data **observed outcomes Y**

Student	Drink	Minutes to finish
Ahmed	energy drink	8
Bader	energy drink	9
Dana	water	11
Reem	water	12

Step 5. The observed difference

The formula applied to the data estimator: difference in means:

```
difference = mean(energy drink) − mean(water)
```

Its value on our data estimate: the number:

```
= (8 + 9) / 2 − (11 + 12) / 2 = 8.5 − 11.5 = −3 minutes
```

The drink arm finished 3 minutes faster on average. Could the coin flips alone produce this?

Step 6. Every other way the coins could have fallen

Under the sharp null each student's time is fixed (Ahmed 8, Bader 9, Dana 11, Reem 12) whatever the drink. The only random thing is which two names the coins picked:

```
number of assignments = C(4, 2) = 4! / (2! × 2!) = 6
```

Compute the difference in means for every assignment:

#	Energy drink	Water	Mean (drink)	Mean (water)	Difference
1	**Ahmed, Bader**	**Dana, Reem**	**8.5**	**11.5**	**−3**
2	Ahmed, Dana	Bader, Reem	9.5	10.5	−1
3	Ahmed, Reem	Bader, Dana	10.0	10.0	0
4	Bader, Dana	Ahmed, Reem	10.0	10.0	0
5	Bader, Reem	Ahmed, Dana	10.5	9.5	+1
6	Dana, Reem	Ahmed, Bader	11.5	8.5	+3

Row 1 (shaded) is what happened. The other rows are what the null would have produced with different coin flips.

Step 7. The p-value randomization test

The p-value is the share of assignments at least as extreme as the observed one, that is a gap of 3 or more in either direction:

```
p = (number of assignments with |difference| ≥ 3) / (total assignments)
```

```
p = 2 / 6 = 0.33
```

Rows 1 and 6 qualify. One-sided ("faster" only): row 1 alone, p = 1/6 = 0.17.

Step 8. The conclusion

Under "the drink does nothing", coin flips alone give a gap this large one time in three. At the usual cut-off of 0.05 we **cannot reject the sharp null**.

This does not show the drink does nothing; four students are too few to tell. The smallest p four students can give is 1/6 (one-sided) or 2/6 (two-sided), so 0.05 is out of reach.

Step 9. What would change the answer

More students. 8 students, 4 treated: C(8, 4) = 70 assignments, smallest two-sided p = 2/70 = 0.03. 20 students, 10 treated: 184,756 assignments. Real experiments sample a few thousand assignments at random (a permutation test); Lab 1 does this with `randomization_test` and 1,000 permutations.

The four words in this example

Word	In this example
Estimand	ATE: the average effect of the drink on finishing time over the four students. Defined before any data.
Estimator	Difference in means: mean time with the drink minus mean time with water.
Estimate	−3 minutes, the number the estimator gave on our data.
Sharp null	$\tau_i = 0$ for every student. Under it the ATE is 0 and every assignment in the table is equally likely.

Three things to remember

1. The sharp null fixes every outcome; the only randomness is the assignment, so no distribution assumptions are needed.

2. The p-value is a count: assignments at least as extreme as ours, divided by all assignments.

3. "Cannot reject" is not "no effect". It is "not enough evidence at this sample size".